

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CONCEPT MAP

Course Code:	ITA0302	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Concept Map
Weightage:	60 % of CIA	Total Marks:	20 marks
CO1 Statement:	Analyse the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.		
Student Name:	Nissanki Sri Venkata Lakshmi Kavya	Reg. No.:	192373036
Section / Batch:	Slot D	Date:	16-07-2026

Code of Conduct

I Nissanki Sri Venkata Lakshmi Kavya(192373036) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.S.V.L.Kavya

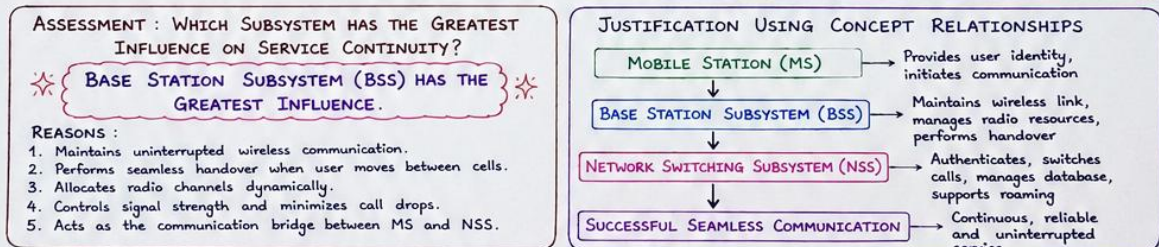
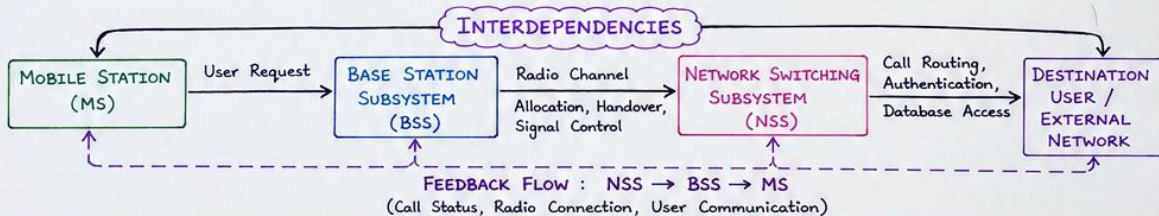
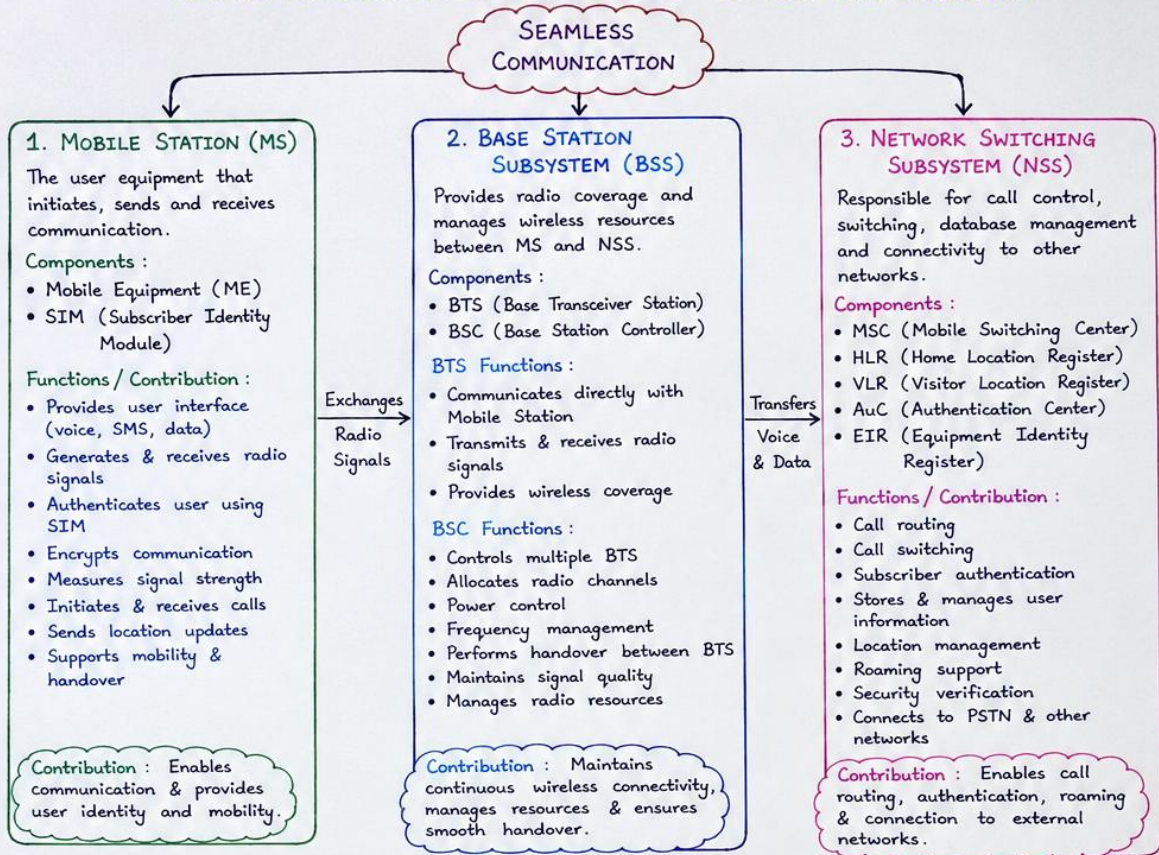
Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Annexure A - QUESTION

1. Construct a concept map that evaluates the contribution of Mobile Station, Base Station Subsystem, and Network Switching Subsystem toward seamless communication.
Illustrate their interdependencies.
Assess which subsystem has the greatest influence on service continuity.
Justify your conclusions using concept relationships.

CONCEPT MAP : EVALUATING THE CONTRIBUTION OF MOBILE STATION, BASE STATION SUBSYSTEM & NETWORK SWITCHING SUBSYSTEM TOWARD SEAMLESS COMMUNICATION



FINAL CONCLUSION

All three subsystems work together for seamless communication. Mobile Station starts the communication, Base Station Subsystem ensures continuous wireless connectivity, and Network Switching Subsystem handles switching, authentication, routing and roaming. However, Base Station Subsystem (BSS) contributes the most to service continuity because it manages radio communication, signal quality and handover, which are essential for uninterrupted mobile communication.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand